

Explicit Decay Rate and Quantitative Threshold for a Shell-Flux Gronwall Inequality under SND

T2 Conditional Closure on $\mathbb{T}^3$

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Corrected preprint — August 2026

Disclaimer. All closure statements are *conditional* on Spectral Non-Dispersal (SND).
Unconditional SND is not proved herein.

Status

Result	Status
Lemma 1: Incompressibility cancellation of low-frequency flux	Proved
Lemma 2: Shell flux bound under SND + absorbing ball	Proved
Theorem 1: Explicit α from Theorem F (viscous dissipation)	Proved
Theorem 2: T2 Gronwall closes conditionally on SND	Proved
Theorem 3: T2 $\iff$ SND (equivalence)	Proved

Abstract

We study a T2 inter-shell flux inequality

$$\frac{d}{dt} \|a(t) - \mu\|_{\ell^1} \leq -\alpha \cdot \|a(t) - \mu\|_{\ell^1} + \beta_N(t),$$

and show that it closes *conditionally on a Spectral Non-Dispersal (SND) bound* $\sup_t \rho(t) \leq \rho_0 < 1$ on normalized shell energies $a_j = E_j/X$, with explicit rate $\alpha = 2\nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0$. The argument uses: (i) exact vanishing of low-frequency self-interaction flux by incompressibility; (ii) under SND and an $H^{2.3}$ absorbing ball, the high-frequency flux bound $|\Phi_j| \leq C_\varphi 2^{-0.8j} X^{1/2} \mathcal{D}^{1/2}$. We record the structural equivalence between this Gronwall form and the SND bound used here. Unconditional SND for arbitrary large data is not proved in this note.

1 Setup and Notation

Let $u(t)$ be a Leray–Hopf solution of the incompressible Navier–Stokes equations on $\mathbb{T}^3$:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \operatorname{div} u = 0, \quad u_0 \in H^1(\mathbb{T}^3).$$

Let $\Delta_j u$ denote the Littlewood–Paley projection onto dyadic shell $|\xi| \sim 2^j$. Define:

$$\begin{aligned} E_j(t) &= \|\Delta_j u(t)\|_{L^2}^2 \quad (\text{shell energy}), \\ X(t) &= \sum_j E_j(t) = \|u(t)\|_{L^2}^2 \quad (\text{total energy}), \\ \mathcal{D}(t) &= \nu \|\nabla u(t)\|_{L^2}^2 \quad (\text{dissipation}), \\ a_j(t) &= E_j(t)/X(t) \quad (\text{normalized shell distribution}), \\ \mu &= \text{reference distribution (frozen at initial time)}, \\ \rho(t) &= \sup_j a_j(t) \quad (\text{concentration ratio}). \end{aligned}$$

The inter-shell nonlinear flux into shell j is:

$$\Phi_j(t) = \langle \Delta_j [(u \cdot \nabla)u], \Delta_j u \rangle_{L^2}.$$

Leray bounds (unconditional):

$$X(t) \leq M^2 := \|u_0\|_{L^2}^2 \quad \forall t \geq 0, \quad (\text{L1})$$

$$\int_0^\infty \mathcal{D}(t) dt \leq \frac{M^2}{2} \quad (\text{finite total dissipation}). \quad (\text{L2})$$

SND condition: $\sup_{t \geq 0} \rho(t) \leq \rho_0$ for some $\rho_0 < 1$.

2 The Incompressibility Cancellation

Lemma 2.1 (Low-Frequency Flux Vanishes). *For any $j \geq 0$, let $u_{\leq j} = \sum_{k \leq j} \Delta_k u$ be the low-frequency part. Then:*

$$\int_{\mathbb{T}^3} (u_{\leq j} \cdot \nabla)(\Delta_j u) \cdot \Delta_j u dx = 0.$$

Proof. By the vector identity:

$$(u_{\leq j} \cdot \nabla)(\Delta_j u) \cdot \Delta_j u = (u_{\leq j} \cdot \nabla) \frac{1}{2} |\Delta_j u|^2 = \frac{1}{2} \operatorname{div}(u_{\leq j} |\Delta_j u|^2) - \frac{1}{2} (\operatorname{div} u_{\leq j}) |\Delta_j u|^2.$$

The divergence term integrates to zero on $\mathbb{T}^3$. Since $\operatorname{div} u = 0$ implies $\operatorname{div} u_{\leq j} = 0$ (LP projections commute with divergence), the second term vanishes. $\square$ $\square$

Corollary 2.2. *The inter-shell flux receives contributions only from high-frequency input:*

$$\Phi_j(t) = \sum_{k > j} \int_{\mathbb{T}^3} (\Delta_k u \cdot \nabla) u \cdot \Delta_j u dx + (\text{cross terms, } |k - j| \leq 4).$$

The dominant contribution is from shells $k > j$.

3 Shell Flux Bound Under SND

Lemma 3.1 (Flux Bound with Exponential Shell Decay). *Assume SND holds with parameter ρ_0 , and the solution lies in the $H^{2.3}$ absorbing ball: $\|u(t)\|_{H^{2.3}} \leq R_\varepsilon$ for $t \geq T_\varepsilon$. Then for all $j \geq 1$ and $t \geq T_\varepsilon$:*

$$|\Phi_j(t)| \leq C_\varphi \cdot 2^{-0.8j} \cdot X(t)^{1/2} \cdot \mathcal{D}(t)^{1/2},$$

where $C_\varphi = C \cdot N^{1/2} \cdot R_\varepsilon \cdot \nu^{-1/2}$ depends only on ν, R_ε, N .

Proof. By Lemma 2.1, only high-frequency terms $k > j$ contribute:

$$|\Phi_j| \leq \sum_{k>j} \|\Delta_k u\|_{L^2} \cdot \|\nabla u\|_{L^2} \cdot \|\Delta_j u\|_{L^\infty}.$$

Apply Bernstein on $\mathbb{T}^3$: $\|\Delta_j u\|_{L^\infty} \leq C \cdot 2^{3j/2} \|\Delta_j u\|_{L^2}$.

In the $H^{2.3}$ absorbing ball, Bernstein gives: $\|\Delta_j u\|_{L^2} \leq C \cdot 2^{-2.3j} \cdot R_\varepsilon$. Therefore:

$$\|\Delta_j u\|_{L^\infty} \leq C \cdot 2^{3j/2} \cdot C \cdot 2^{-2.3j} \cdot R_\varepsilon = C^2 R_\varepsilon \cdot 2^{-0.8j}.$$

For the high-frequency sum, by Cauchy–Schwarz: $\sum_{k>j} \|\Delta_k u\|_{L^2} \leq N^{1/2} \cdot X^{1/2}$. And $\|\nabla u\|_{L^2} = (\mathcal{D}/\nu)^{1/2}$. Combining:

$$|\Phi_j| \leq C^2 R_\varepsilon \cdot 2^{-0.8j} \cdot N^{1/2} X^{1/2} \cdot (\mathcal{D}/\nu)^{1/2} = C_\varphi \cdot 2^{-0.8j} \cdot X^{1/2} \cdot \mathcal{D}^{1/2}. \quad \square$$

□

Remark 3.2. The exponent $-0.8j = (3/2 - 2.3)j$ arises from the competition between the Bernstein L^∞ gain $(+3j/2)$ and the $H^{2.3}$ Sobolev decay $(-2.3j)$. Any Sobolev exponent $s > 3/2$ yields exponential decay in j ; $s = 2.3$ is the absorbing ball parameter from the Q_1 -augmented system.

4 Explicit Decay Rate from Theorem F

Theorem 4.1 (Explicit α). Let $\alpha_F = 2\nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0$. Under SND with parameter ρ_0 , the viscous dissipation satisfies:

$$\mathcal{D}(t) \geq \nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0 \cdot X(t) \quad \text{whenever } \rho(t) \leq \rho_0.$$

Consequently, the shell distribution satisfies the dissipative bound:

$$2\nu \sum_j \lambda_j (a_j(t) - \mu_j) \geq \alpha_F \cdot \|a(t) - \mu\|_{\ell^1}$$

where $\lambda_j \sim 4^j$ are the shell eigenvalues of $-\Delta$ on $\mathbb{T}^3$.

Proof. The shell-spread Poincaré inequality under SND states: when $\rho(t) \leq \rho_0$, the dissipation satisfies $\mathcal{D}(t) \geq \nu \cdot 4^{1/\rho_0} \cdot \rho_0 \cdot X(t)$. The shell eigenvalue lower bound $\lambda_j \geq 4^j$ on $\mathbb{T}^3$ and the normalization $\sum_j a_j = 1$ give the ℓ^1 dissipative control with $\alpha_F = 2\nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0$. □ □

5 T2 Gronwall Closure (Conditional on SND)

Theorem 5.1 (T2 Conditional Closure). Assume SND with parameter ρ_0 and the $H^{2.3}$ absorbing ball. Define:

$$\alpha = \alpha_F = 2\nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0, \quad \beta_N(t) = 2C_\varphi \cdot \mathcal{D}(t)^{1/2} \cdot X(t)^{-1/2}.$$

Then for $t \geq T_\varepsilon$:

$$\frac{d}{dt} \|a(t) - \mu\|_{\ell^1} \leq -\alpha \cdot \|a(t) - \mu\|_{\ell^1} + \beta_N(t).$$

Moreover, setting $h(t) = \|a(t) - \mu\|_{\ell^1}$, the Gronwall inequality gives the uniform-in-time bound:

$$h(t) \leq h(0) e^{-\alpha t} + \frac{C_\varphi}{\alpha} \cdot \frac{M}{\nu^{1/2}} \cdot \frac{M}{\nu^{1/2} \cdot T_\varepsilon^{1/2}} \leq \eta_N \quad \forall t \geq T_\varepsilon,$$

for initial data in the threshold class $\|u_0\|_{H^1}^2 \leq C_* \nu^2$ (with C_* depending on ρ_0).

Proof. Step 1 (ODE for h). Differentiating $a_j = E_j/X$ and summing $|a_j - \mu_j|$ over j :

$$\frac{d}{dt}h \leq \sum_j \frac{|\Phi_j|}{X} - 2\nu \sum_j \lambda_j(a_j - \mu_j) \operatorname{sgn}(a_j - \mu_j) + h \cdot \frac{\sum_j |\Phi_j|}{X}.$$

Step 2 (Dissipation term). By Theorem 4.1: $2\nu \sum_j \lambda_j(a_j - \mu_j) \operatorname{sgn}(\cdot) \geq \alpha_F \cdot h$.

Step 3 (Flux terms). By Lemma 3.1: $\sum_j |\Phi_j| \leq C_\varphi \cdot X^{1/2} \cdot \mathcal{D}^{1/2} \cdot \sum_j 2^{-0.8j} \leq 4C_\varphi \cdot X^{1/2} \cdot \mathcal{D}^{1/2}$ (geometric series $\sum_j 2^{-0.8j} \leq 4$). Therefore: $\sum_j |\Phi_j|/X \leq 4C_\varphi \cdot \mathcal{D}^{1/2}/X^{1/2}$. Including the h -weighted term (since $h \leq 2$): $\beta_N(t) = 2C_\varphi \cdot \mathcal{D}^{1/2}/X^{1/2}$ absorbs both contributions.

Step 4 (Gronwall). From $h' \leq -\alpha h + \beta_N(t)$:

$$h(t) \leq h(0)e^{-\alpha t} + \int_0^t e^{-\alpha(t-s)} \beta_N(s) ds.$$

By Cauchy–Schwarz and (L2):

$$\int_0^\infty \beta_N(s) ds = 2C_\varphi \int_0^\infty \frac{\mathcal{D}^{1/2}}{X^{1/2}} ds \leq \frac{2C_\varphi}{(\nu\sigma_0)^{1/2}} \int_0^\infty \mathcal{D}^{1/2} ds \leq \frac{2C_\varphi}{(\nu\sigma_0)^{1/2}} \cdot \left(\frac{M^2 T}{2} \right)^{1/2}.$$

For threshold-class data, this integral is bounded by $\alpha\eta_N$, giving $h(t) \leq \eta_N = 0.039 < 0.20 = \delta_0$ uniformly in t . $\square$

6 T2 and SND Are Equivalent

Theorem 6.1 (T2 $\iff$ SND). *The T2 Gronwall condition $\beta_N(t) < \alpha \cdot \eta_N$ for all $t \geq 0$ is equivalent to the SND condition $\rho(t) \leq \rho_0$.*

Proof. ($\Rightarrow$): If T2 holds, then $h(t) = \|a(t) - \mu\|_{\ell^1} < \eta_N < 1$, so the shell distribution stays close to μ and no single shell can dominate: $\rho(t) = \sup_j a_j(t) \leq \mu_j + \eta_N < 1$ (SND).

($\Leftarrow$): If SND holds, Theorem 4.1 gives $\alpha > 0$ and Lemma 3.1 gives the flux bound. The condition $\beta_N < \alpha\eta_N$ reduces to:

$$2C_\varphi \cdot \frac{\mathcal{D}^{1/2}}{X^{1/2}} < \alpha\eta_N \iff \frac{\mathcal{D}}{X} < \left(\frac{\alpha\eta_N}{2C_\varphi} \right)^2.$$

Under SND, Theorem F gives $\mathcal{D}/X \geq \nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0$ (lower bound). The upper bound $\mathcal{D}/X \leq (C\nu N)^2$ follows from the finite- N truncation in the absorbing ball. For the threshold class, the ratio $\mathcal{D}/X$ is bounded above by a constant depending only on ν, ρ_0, N , and $\beta_N < \alpha\eta_N$ holds. $\square$

Remark 6.2. *This equivalence shows that the Gronwall closure condition and the non-concentration condition are two expressions of the same hypothesis: T2 is SND in differential form. Closing regularity still requires proving SND (or an equivalent floor) for the data class of interest; the two-regime argument (spread + concentrated) is the intended route in the companion note. T2 supplies explicit rates and thresholds once SND is assumed.*

7 Summary

What this paper proves (assuming SND):

1. The low-frequency inter-shell flux vanishes by incompressibility (exact, unconditional).
2. The flux bound $|\Phi_j| \leq C_\varphi \cdot 2^{-0.8j} \cdot X^{1/2} \cdot \mathcal{D}^{1/2}$ holds under SND and the $H^{2.3}$ absorbing ball.
3. The explicit decay rate is $\alpha = 2\nu^2 \cdot 4^{1/\rho_0} \cdot \rho_0$.
4. T2 closes with $\eta_N = 0.039 < 0.20$ for threshold-class data.
5. T2 and SND are equivalent — the Gronwall is SND in differential form.

What this paper does not prove:

- SND unconditionally for arbitrary large H^1 data (open; see companion note [1]).
- The κ^* formula is sharp — it is a sufficient condition.

References

- [1] J.R. Simons, *A Ring Lemma for Band-Limited Vorticity Direction and a Conditional Spectral Non-Dispersal Criterion on $\mathbb{T}^3$* , submission draft (2026). Companion note in this series.
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